WEEK 4 · LESSON 21 OF 24 · HANDOUT

Derivatives: The Maths of Change

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=9vKqVkMQHKk — open the lesson online to watch inside the course.

🎓 **Go deeper:** 3Blue1Brown: The Essence of Calculus — https://www.youtube.com/watch?v=WUvTyaaNkzM

Your goal in this lesson: Explain a derivative as a slope or rate of change, calculate simple derivatives by rule and by computer, and use derivatives to find where a function is lowest.

Why AI needs calculus

Training a model means **making its error as small as possible**. To do that, the model must know: *if I change this weight a little, does the error go up or down — and how fast?* That question is answered by a **derivative**.

Slope: change in output per change in input

For a straight line, the **slope** is constant:

> *slope = change in y ÷ change in x*

A moto fare of 400 + 230 × km has slope **230**: each extra kilometre adds 230 RWF.

For a curve, the slope is different at every point. The **derivative** f′(x) is the slope at one exact point — the slope of the **tangent line** that just touches the curve there.

Rate of change in daily life

- Speed is the derivative of distance with respect to time.
- If a bus has travelled d(t) km after t hours, d′(2) is its speed at exactly 2 hours.
- Marginal cost in business is the derivative of total cost with respect to quantity.

Calculating a derivative with a computer

Take a tiny step h and measure the slope:

> $f'(x) \approx (f(x + h) - f(x)) \div h$

```
def f(x):
    return x ** 2

def derivative(func, x, h=1e-6):
    return (func(x + h) - func(x)) / h

for x in [-2, 0, 1, 3]:
    print(x, round(derivative(f, x), 3))
```

The results are close to −4, 0, 2 and 6 — always **2x**.

A few rules

Function	Derivative	Example
c (a constant)	0	f(x) = 5 → 0
c × x	c	f(x) = 230x → 230
x^2	2x	at x = 3, slope 6
x^n	$n \times x^{n-1}$	$x^3 \to 3x^2$
$(x - a)^2$	2(x − a)	used for squared error
sum of functions	sum of derivatives	$x^2 + 3x \to 2x + 3$

You do not need to memorise many rules for AI — libraries such as PyTorch calculate derivatives automatically (**autodiff**). But you must understand what they mean.

The lowest point has slope zero

At the bottom of a valley the ground is flat: **slope = 0**. To the left the slope is negative (going down), to the right positive (going up).

```
import numpy as np
import matplotlib.pyplot as plt

def error(w):
    return (w - 3) ** 2 + 2

ws = np.linspace(-1, 7, 200)
plt.plot(ws, error(ws))
plt.axvline(3, linestyle="--")
plt.title("Error is lowest where the slope is 0 (w = 3)")
plt.xlabel("weight w")
plt.ylabel("error")
plt.show()

for w in [1, 3, 5]:
    print(w, round(derivative(error, w), 2))   # negative, ~0, positive
```

The derivative tells you **which way is downhill**: if the slope is positive, decrease w; if negative, increase w. That is the whole idea of gradient descent — next lesson.

More than one input: partial derivatives and the gradient

A model has many weights. A **partial derivative** measures the slope when you change **one** weight and keep the others fixed. All the partial derivatives together form the **gradient** — an arrow pointing in the direction where the error rises fastest.

```
def error2(w, b):
    return (w - 3) ** 2 + (b + 1) ** 2

w, b, h = 0.0, 0.0, 1e-6
dw = (error2(w + h, b) - error2(w, b)) / h
db = (error2(w, b + h) - error2(w, b)) / h
print(round(dw, 2), round(db, 2))   # about -6 and 2
```

Key terms

- **Slope** — change in output per unit change in input.
- **Derivative** — the slope at one point; the instant rate of change.
- **Tangent line** — the line that touches a curve at one point with the same slope.
- **Partial derivative** — the slope with respect to one input, others fixed.
- **Gradient** — all partial derivatives together; points uphill.
- **Autodiff** — software that computes derivatives automatically.

Remember this: A derivative is a slope: how fast the output changes when the input changes. The minimum of a function is where the slope is zero, and the sign of the slope tells you which way is downhill.

Name: _____ Date: _____

Activity

Task 1

Use `derivative()` on $f(x) = x^3$ at $x = 2$. Check with the rule $3x^2$.

Task 2

A shop's profit is $p(q) = -2q^2 + 400q$ (q = units). Find the slope at $q = 50$, 100 and 150. Where is profit highest?

Task 3

Plot `error2` for $b = -1$ as w changes. Where is the lowest point?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does a derivative measure?
   - A. The area under a curve
   - B. The slope or instant rate of change at a point
   - C. The number of inputs
   - D. The average of the data
2. What is the derivative of $f(x) = x^2$ at $x = 3$?
   - A. 9
   - B. 3
   - C. 2
   - D. 6

3. A fare is 400 + 230 × km. What is the derivative with respect to km?

A. 230
B. 400
C. 0
D. 630

4. At the lowest point of a smooth error curve, the derivative is…

A. Zero
B. Undefined always
C. Very large
D. Negative

5. At the current weight the derivative of the error is positive. To reduce the error you should…

A. Double the error
B. Not change anything
C. Increase the weight
D. Decrease the weight

6. What is the gradient?

A. All the partial derivatives together, pointing uphill
B. The data matrix
C. A colour fade
D. A chart title

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